Knowledge Graph for Graph Data Integration Research

Project proposal and implementation plan. This project will study how research papers describe graph-based data integration. Task 1 will build a metadata graph from OpenAlex records. Task 2 will extend that graph with facts extracted from available abstracts, preserve their evidence, and evaluate extraction and entity matching.

1. Problem and scope

Data engineering students comparing integration approaches need to connect publications with their authors, venues, methods, technologies, and the problems those approaches address. A publication list supports basic bibliographic questions; a knowledge graph can connect these facts across papers and make the evidence traceable to its source.

The project will create an exploratory literature graph, not a ranking of technologies or a claim that one approach is universally better. Task 1 will query OpenAlex with the phrase “graph data integration” for publication years 2018-2025. This bounded search is not an exhaustive review and will not be presented as a language-filtered sample. Task 2 will use eligible abstracts available for the collected works, record exclusions, and will not ingest full papers or infer causal performance claims.

2. Data sources and access

The OpenAlex Works API will supply structured publication metadata and, where available, abstract inverted indexes. A command-line fetcher will reconstruct abstracts and save them separately. OpenAlex work ID will connect the metadata and text records; DOI will be retained as a secondary identifier when available. The retrieval log will record the search phrase, year range, limit, date, and returned counts.

Planned input	Planned contents and role
works.csv	OpenAlex work ID, DOI, title, publication year, venue, authors and IDs, source URL, work type.
abstracts.csv	Available abstracts keyed by OpenAlex work ID; some collected works may lack an abstract. Used as Task 2 text fields.

Task 1 will report counts of Work, Person, and Venue nodes, relationships, rejected rows, and year coverage so the initial graph can be checked and reproduced.

3. Graph schema and identity rules

Task 1 will build the initial schema below. Task 2 will extend the same graph; publication year and other descriptive values will remain properties rather than separate nodes.

(Work)-[AUTHORED_BY]->(Person) (Work)-[PUBLISHED_IN]->(Venue)

(Work)-[:DISCUSSES]->(Method | Technology | DataProblem)

(Method)-[:ADDRESSES]->(DataProblem)

(Method)-[:USES]->(Technology)

Stable identifiers. Work will use its OpenAlex ID, with DOI as fallback. Person and Venue will use OpenAlex IDs; where an ID is unavailable, a normalized name may be used provisionally and flagged for review. Task 2 concepts will use a canonical label and controlled aliases. Names alone will not be treated as proof of identity; uncertain matches will remain unresolved. Each extracted fact will retain its source work, supporting passage, extraction method, confidence, and whether the relation is stated or inferred. Co-occurrence alone will not create a relationship.

Sample record and text passage

Candidate schema-pilot record: “Evaluation of Data Integration Plans based on Graph Data” (2021), DOI 10.1016/j.procs.2021.08.107, published in Procedia Computer Science; OpenAlex work ID W3204151456. Verify the metadata and abstract against OpenAlex during collection.

Abstract passage, paraphrased for the schema pilot: the paper contrasts RDF graphs created from non-graph sources with GraphQL used as a proxy for virtual access to heterogeneous sources. This passage supports stated mentions and relations; it does not by itself support a performance ranking.

Source fact	**Graph representation**	**Evidence / check**
The work has DOI 10.1016/j.procs.2021.08.107.	Work.doi	Structured OpenAlex record; compare DOI and source URL.
The work appeared in 2021.	Work.year = 2021	Structured metadata; compare the record.
The work was published in Procedia Computer Science.	Work -[:PUBLISHED_IN]-> Venue	Structured metadata; check venue name and ID.
The abstract discusses RDF graphs built from non-graph data sources.	Work -[:DISCUSSES]-> Technology(RDF)	Abstract passage; record the exact supporting span.
The abstract describes GraphQL as a proxy for virtual access to heterogeneous sources.	Work -[:DISCUSSES]-> Technology(GraphQL); Work -[:DISCUSSES]-> DataProblem(heterogeneous-source integration)	Abstract passage; mark only directly supported facts as stated.

4. Questions and planned checks

Task 1 will answer Q1-Q4 using the structured graph. Q4 is the required multi-relationship query and will be implemented as a runnable Cypher query with saved results. Task 2 will add Q5-Q6 and revisit all five questions to compare answers before and after text enrichment.

Question	Expected answer and graph path	Check
Q1 Which works in the collected sample were authored by Ana-Maria Ghiran?	Work -AUTHORED_BY-> Person. Return title, DOI, and year.	Check returned author ID and each work against the OpenAlex row.
Q2 How many works are recorded for each publication year?	Count Work nodes by year property.	Reconcile with the frozen input and build statistics; explain the search-sample limits.
Q3 How many works are linked to each venue?	Work -PUBLISHED_IN-> Venue.	Compare counts with structured records; inspect missing venue values.
Q4 Which different authors are connected through separate works published in the same venue?[complex]	Person <-AUTHORED_BY- Work -PUBLISHED_IN-> Venue <-PUBLISHED_IN- Work -AUTHORED_BY-> Person.	Check author and venue IDs, distinct works, and each path against the records. Avoid describing this venue link as collaboration unless the records support that claim.
Q5 For a selected author, which of their papers discuss a method for addressing heterogeneous-source integration problems, and what venue/year are recorded?	Person <-AUTHORED_BY- Work -DISCUSSES-> Method -ADDRESSES-> DataProblem; also Work -PUBLISHED_IN-> Venue.	Trace every text-derived edge to a passage; check author, venue, and year against structured metadata. Label stated and inferred evidence separately.
Q6 Which integration methods are explicitly described as using RDF or GraphQL, and what data integration problems do they address?	Work → DISCUSSES → Method → USES → Technology (RDF or GraphQL) Method → ADDRESSES → DataProblem	Manually trace each USES and ADDRESSES relationship to the supporting abstract passages. Verify that relationships are explicitly stated rather than inferred from simple co-occurrence. Check the source work ID, passage ID, and evidence status for each relationship.

This allocation follows the assignment: Task 1 must answer 1 - 4 questions; Task 2 must revisit all six, including information with a text-derived fact.

5. Task 1 implementation and checks

The Task 1 script will retrieve and freeze structured records, construct the Work, Person, and Venue nodes and their relationships, and export the graph and Neo4j import files. It will reject and log work rows without an OpenAlex ID or DOI, title, or valid year; treat DOI and venue as optional; deduplicate nodes and edges by stable IDs; and flag conflicting properties for review. The Neo4j import will use uniqueness constraints and MERGE so repeated imports do not create duplicate nodes or relationships. Validation will check required properties, identifier uniqueness, allowed relationship types, and that each relationship connects existing nodes of the expected types. The work will report graph statistics and manually compare a sample of facts with OpenAlex records.

Task 1 deliverables will include at least three runnable queries with saved results and explanations, including Q4 as a multi-hop query; a validation report; a sample source check; an initial graph export; and a README with software requirements and rebuild steps. The Neo4j database will be checked with counts and query results before the defense.

6. Task 2 work plan and evaluation

Text preparation and extraction. Freeze the abstract export produced for Task 1, select a representative sample across available years and, where possible, venues. Preserve source work IDs. Clean only formatting needed for processing and split abstracts into sentences or short passages without changing meaning. Extract Method, Technology, and DataProblem mentions and typed relationships using a constrained LLM prompt, rules, or both. Store raw outputs before correction; validate schema, allowed types, endpoints, and passage support; retain rejected and uncertain results.

Entity matching and graph update. Standardize concept labels with a documented alias list and context. Link extracted facts to Work nodes created in Task 1 by OpenAlex ID; resolve concepts by canonical label and context. Keep different entities separate when labels are identical but refer to different objects; leave uncertain matches unresolved. Report examples of successful merges, separate same-name entities, and unresolved matches where present, and explain any absent cases. Add accepted facts to the Task 1 graph, with a passage pointer and provenance on every text-derived relationship. Preserve the Task 1 graph export for comparison.

Evaluation. Hold out the agreed passage sample from prompt examples and rule tuning. A human annotator will mark all in-scope entity mentions, identities, relationship labels, directions, endpoints, and supporting spans. Run a fixed baseline and one justified change, such as a controlled alias list, on the same sample with identical scoring rules. Report entity and relationship precision, recall, and F1 with correct, incorrect, and missed counts; report entity matching as correct matches / total checked plus unresolved cases. State the cases included in each calculation. Preserve uncorrected outputs, separate any post-correction results, and discuss representative errors and their effects on graph answers. No accuracy threshold is assumed.

Tools and fallback. Record the exact LLM/provider and model version, run date, settings, prompts/rules, and outputs used.

7. Schedule and submission

Milestone	Work and evidence
Proposal checkpoint - 3 Nov	Submit the bounded scope, sample mapping, schema, five questions with task assignments, checks, and fallback.
Task 1 - by 1 Dec	Build and validate the initial graph; run Q1-Q4, save results and explanations, manually check a sample against sources, and prepare the README rebuild steps and graph export.
Task 2 - December credit day	Add text facts to the same graph; preserve prompts/rules and exact outputs, reference annotations, metrics, error analysis, and before/after answers for all five questions.